

```
# VENKATESH GAURI SHANKAR
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```
#To implement association rule mining for various datasets.
```

```
pip install apyori
```

```
Requirement already satisfied: apyori in /usr/local/lib/python3.7/dist-packages (1.1.2)
```

```
#Read data and Display
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from apyori import apriori
```

```
store_data = pd.read_csv("store_data.csv", header=None)
display(store_data.head())
print(store_data.shape)
```



|   | 0             | 1         | 2          | 3                | 4            | 5                | 6    | 7              | 8            | 9            | 10             | 11        |
|---|---------------|-----------|------------|------------------|--------------|------------------|------|----------------|--------------|--------------|----------------|-----------|
| 0 | shrimp        | almonds   | avocado    | vegetables mix   | green grapes | whole weat flour | yams | cottage cheese | energy drink | tomato juice | low fat yogurt | green tea |
| 1 | burgers       | meatballs | eggs       | NaN              | NaN          | NaN              | NaN  | NaN            | NaN          | NaN          | NaN            | NaN       |
| 2 | chutney       | NaN       | NaN        | NaN              | NaN          | NaN              | NaN  | NaN            | NaN          | NaN          | NaN            | NaN       |
| 3 | turkey        | avocado   | NaN        | NaN              | NaN          | NaN              | NaN  | NaN            | NaN          | NaN          | NaN            | NaN       |
| 4 | mineral water | milk      | energy bar | whole wheat rice | green tea    | NaN              | NaN  | NaN            | NaN          | NaN          | NaN            | NaN       |

(7501, 20)

```
#Preprocessing on Data
```

```
records = []
for i in range(1, 7501):
    records.append([str(store_data.values[i, j]) for j in range(0, 20)])
```

```
print(type(records))
```

```
<class 'list'>
```

```
# Apriori
```

```
association_rules = apriori(records, min_support=0.0045, min_confidence=0.2, min_lift=3, min_length=2)
association_results = list(association_rules)
```

```
print("There are {} Relation derived.".format(len(association_results)))
```

There are 48 Relation derived.

```
for i in range(0, len(association_results)):
    print(association_results[i][0])
```

```
frozenset({'light cream', 'chicken'})
frozenset({'mushroom cream sauce', 'escalope'})
frozenset({'escalope', 'pasta'})
frozenset({'ground beef', 'herb & pepper'})
frozenset({'ground beef', 'tomato sauce'})
frozenset({'olive oil', 'whole wheat pasta'})
frozenset({'shrimp', 'pasta'})
frozenset({'light cream', 'nan', 'chicken'})
frozenset({'frozen vegetables', 'chocolate', 'shrimp'})
frozenset({'ground beef', 'cooking oil', 'spaghetti'})
frozenset({'mushroom cream sauce', 'nan', 'escalope'})
frozenset({'nan', 'escalope', 'pasta'})
frozenset({'ground beef', 'frozen vegetables', 'spaghetti'})
frozenset({'olive oil', 'milk', 'frozen vegetables'})
frozenset({'mineral water', 'frozen vegetables', 'shrimp'})
frozenset({'olive oil', 'frozen vegetables', 'spaghetti'})
frozenset({'frozen vegetables', 'shrimp', 'spaghetti'})
frozenset({'tomatoes', 'frozen vegetables', 'spaghetti'})
frozenset({'ground beef', 'grated cheese', 'spaghetti'})
frozenset({'ground beef', 'mineral water', 'herb & pepper'})
frozenset({'ground beef', 'nan', 'herb & pepper'})
frozenset({'ground beef', 'spaghetti', 'herb & pepper'})
frozenset({'ground beef', 'milk', 'olive oil'})
frozenset({'ground beef', 'tomato sauce', 'nan'})
frozenset({'ground beef', 'shrimp', 'spaghetti'})
frozenset({'olive oil', 'milk', 'spaghetti'})
frozenset({'mineral water', 'soup', 'olive oil'})
frozenset({'olive oil', 'whole wheat pasta', 'nan'})
frozenset({'nan', 'shrimp', 'pasta'})
frozenset({'olive oil', 'pancakes', 'spaghetti'})
frozenset({'nan', 'frozen vegetables', 'chocolate', 'shrimp'})
frozenset({'ground beef', 'cooking oil', 'nan', 'spaghetti'})
frozenset({'ground beef', 'frozen vegetables', 'nan', 'spaghetti'})
frozenset({'mineral water', 'milk', 'frozen vegetables', 'spaghetti'})
frozenset({'olive oil', 'milk', 'frozen vegetables', 'nan'})
frozenset({'mineral water', 'frozen vegetables', 'nan', 'shrimp'})
frozenset({'olive oil', 'frozen vegetables', 'nan', 'spaghetti'})
frozenset({'spaghetti', 'frozen vegetables', 'nan', 'shrimp'})
frozenset({'tomatoes', 'frozen vegetables', 'nan', 'spaghetti'})
frozenset({'ground beef', 'grated cheese', 'nan', 'spaghetti'})
frozenset({'ground beef', 'mineral water', 'nan', 'herb & pepper'})
frozenset({'ground beef', 'nan', 'spaghetti', 'herb & pepper'})
frozenset({'ground beef', 'milk', 'nan', 'olive oil'})
frozenset({'ground beef', 'spaghetti', 'nan', 'shrimp'})
frozenset({'olive oil', 'milk', 'nan', 'spaghetti'})
frozenset({'mineral water', 'soup', 'nan', 'olive oil'})
frozenset({'olive oil', 'pancakes', 'nan', 'spaghetti'})
frozenset({'nan', 'spaghetti', 'mineral water', 'milk', 'frozen vegetables'})
```

```
for item in association_results:
    # first index of the inner list
    # Contains base item and add item
    pair = item[0]
    items = [x for x in pair]
    print("Rule: " + items[0] + " -> " + items[1])
```

```
# second index of the inner list
print("Support: " + str(item[1]))

# third index of the list located at 0th
# of the third index of the inner list

print("Confidence: " + str(item[2][0][2]))
print("Lift: " + str(item[2][0][3]))
print("=====")
```

```
Rule: light cream -> chicken
Support: 0.004533333333333334
Confidence: 0.2905982905982906
Lift: 4.843304843304844
=====
Rule: mushroom cream sauce -> escalope
Support: 0.005733333333333333
Confidence: 0.30069930069930073
Lift: 3.7903273197390845
=====
Rule: escalope -> pasta
Support: 0.005866666666666667
Confidence: 0.37288135593220345
Lift: 4.700185158809287
=====
Rule: ground beef -> herb & pepper
Support: 0.016
Confidence: 0.3234501347708895
Lift: 3.2915549671393096
=====
Rule: ground beef -> tomato sauce
Support: 0.005333333333333333
Confidence: 0.37735849056603776
Lift: 3.840147461662528
=====
Rule: olive oil -> whole wheat pasta
Support: 0.008
Confidence: 0.2714932126696833
Lift: 4.130221288078346
=====
Rule: shrimp -> pasta
Support: 0.005066666666666666
Confidence: 0.3220338983050848
Lift: 4.514493901473151
=====
Rule: light cream -> nan
Support: 0.004533333333333334
Confidence: 0.2905982905982906
Lift: 4.843304843304844
=====
Rule: frozen vegetables -> chocolate
Support: 0.005333333333333333
Confidence: 0.23255813953488372
Lift: 3.260160834601174
=====
Rule: ground beef -> cooking oil
Support: 0.0048
Confidence: 0.5714285714285714
Lift: 3.281557646029315
=====
Rule: mushroom cream sauce -> nan
Support: 0.005733333333333333
```

Confidence: 0.30069930069930073  
Lift: 3.7903273197390845  
=====  
Rule: nan -> escalope  
Support: 0.005866666666666667  
Confidence: 0.37788135503770315

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